

Revisiting Hall-Petch strengthening in FCC multi-principal element alloys: what grain size, composition, and processing can and cannot explain

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Abstract

Grain size is one of the primary microstructural variables controlling strength, but its role in FCC multi-principal element alloys cannot be separated easily from the effects of chemistry and processing. In this work, we examine the relationships among grain size, composition, processing, yield strength, and Vickers hardness using 94 alloy candidates representing 82 compositions and six experimental batches, spanning grain sizes of 15–212 μm and yield strengths of 152–544 MPa. The classical Hall–Petch relation provides a reproducible yield-strength baseline, giving $R^2 = 0.41$ under record-level and 0.37 under batch-level validation, while the inclusion of composition raises these values to 0.65 and 0.63 with a shared slope of 766 MPa· $\mu\text{m}^{1/2}$. The strongest low-complexity model also contains a coupled contribution from mean grain size and its within-map standard deviation, reaching 0.69 under both protocols. However, the present data cannot separate the physical origin of this contribution from the covariance among composition, processing, and microstructure. The tested solid-solution-strengthening estimates and nonlinear regressors do not improve batch-level predictions once the measured inputs are included. Hardness exhibits a different dependence and does not provide a transferable substitute for yield strength across the campaigns. The comparison with a heterogeneous literature dataset shows that transfer across studies will require prospective datasets in which composition, processing history, microstructure, test conditions, and other experimental factors are documented and controlled consistently.

Keywords: High-entropy alloys, Yield strength, Vickers hardness, Solid solution strengthening, Symbolic regression, Cross-validation

1. Introduction

Grain size is one of the primary microstructural variables controlling strength in FCC alloys, and its contribution is usually described by the Hall-Petch relation

$$\sigma_y = \sigma_0 + k_{\text{HP}}d^{-1/2} \quad (1)$$

which separates a grain-size-independent intercept from the grain-boundary strengthening term [1, 2, 3]. In concentrated FCC alloys, composition can influence solid-solution resistance, ordering, stacking fault energy, and boundary-mediated deformation. Hall-Petch slopes and intercepts therefore vary significantly among alloy systems [4, 5, 6]. Several other grain-size laws can fit a limited range of d almost as well as Eq. (1). A good regression fit alone therefore does not completely and accurately identify the underlying microscopic mechanism.

Multi-principal element alloys (MPEAs) span composition and processing spaces that are high dimensional and too large to explore exhaustively [7, 8, 9, 10]. Closed-loop machine learning (ML) can speed up this search [11, 12, 13, 14]. However, a property model becomes harmful when a good interpolation score is

used to support decisions about new compositions, processing routes, or experimental campaigns. This risk is especially high for small and sparse experimental datasets that represent these very high dimensional design spaces. Nearby compositions, repeated conditions, and campaign-specific processing can make a random train/test split much easier than the intended alloy design problem [15, 16, 17].

Physics-based solid solution strengthening (SSS) models and composition descriptors provide compact alternatives to using unconstrained elemental fractions [18, 19, 20, 21, 22]. However, these descriptors are calculated directly from composition and may add little information after composition is already included. Non-linear and symbolic models can capture interactions that a linear model misses, but they also search a much larger space. As a result, they can exploit campaign identity, correlated descriptors, or unstable denominators. A feature can be physically meaningful without being transferable, and an equation can be interpretable without being a validated physical law.

Hardness serves as an additional, complementary target. Vickers hardness captures the material's resistance to deformation under localized loading. The empirical Tabor relation relates hardness to a representative flow stress, rather than directly to 0.2% offset YS [23, 24]. Applying the same modeling and validation framework to YS and HV shows whether the relevant features depend on the property being modeled.

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Figure 1 and Table 1 define the five model families used throughout this paper, with the same names and order: Family 1 (Classical Hall-Petch), Family 2 (Physics-derived descriptors), Family 3 (Composition/processing), Family 4 (Non-linear estimators), and Family 5 (Symbolic regression). The families ask five sequential questions: What does grain size explain? Do physics-based descriptors capture the remaining strength? What do measured composition and processing add? Does non-linearity help when inputs are matched? Can symbolic search produce useful closed forms? A higher family number does not imply greater physical fidelity. We combine this hierarchy with 5-fold CV, record-level LOO, batch-level LOBO, an external literature stress test, and singularity auditing of closed-form equations.

Table 1: Five model families used in the main paper and Supplementary Information.

Family	Fixed name	Analyses
1	Classical Hall-Petch	Classical and alternative grain-size laws
2	Physics-derived descriptors	SSS models, Wen descriptors, and PCA-OLS
3	Composition/processing	OLS and composition-processing Hall-Petch models, including grain-width interactions
4	Non-linear estimators	Matched linear and non-linear estimators
5	Symbolic regression	PySR, SISSO, and fixed closed forms

2. Dataset and descriptors

2.1. Alloy Synthesis and characterization

The dataset contains 94 FCC alloy conditions in the Al-Co-Cr-Cu-Fe-Mn-Ni-V system. YS is available for 93 conditions, and HV is available for all 94. An earlier Campaign A established the closed-loop, multi-objective alloy-discovery workflow used across this effort [12]; its records are not part of the present property modeling dataset because that campaign used a different processing route. The samples analyzed here were generated from the subsequent B and C Bayesian optimization campaigns at Texas A&M University within BIRDSHOT Center, supported by the U.S. Army DEVCOM High-Throughput Materials Discovery for Extreme Conditions (HTMDEC) initiative [13, 14]. The B campaign contributes 35 candidates from batches BBA, BBB, and BBC, including 34 with YS. It explored composition under an FCC phase stability constraint while holding processing fixed. The complete B-campaign synthesis and specimen preparation workflow is documented in Ref. [14].

The C campaign contributes 59 candidates from CBA, CBB, and CBC. Building on the B-campaign results, it focused on a V-Cr-Mn-Fe-Co-Ni design space and jointly varied composition with three processing variables: cold work, recrystallization temperature, and hold time. Candidate alloys were iteratively selected from an FCC-constrained, manufacturable design space with the models updated after each iteration using the newly acquired data. A modified JMAK-type static-recrystallization and grain-growth prior supplied grain-size estimates over the feasible design space to guide candidate selection. That model

and the complete C-campaign workflow will be reported in a separate future publication. The property models evaluated here use grain sizes measured by electron backscatter diffraction (EBSD), not these design-stage predictions.

Overall, both campaigns used the same high level production route: high-purity elemental feedstock was vacuum arc melted, homogenized, cold rolled, recrystallized, and quenched. Energy-dispersive X-ray spectroscopy (EDS) and X-ray diffraction were used to check realized chemistry and phase constitution, while electron microscopy and EBSD documented the microstructure. Grain size was measured from EBSD maps on surfaces normal to the rolling direction and summarized by the equivalent circle mean d and within-map standard deviation SD_{grain} . Representative candidate alloy verification is shown in Figure 2. Quasistatic tensile testing was used to determine 0.2% offset YS, while Vickers indentation tests provided the HV. Further details for BBA-BBC are given in Ref. [14]; a detailed account of CBA-CBC will follow in the campaign publication.

2.2. Coverage, dependence, and feature sets

The 94 candidates in this study represent 82 unique compositions: 73 chemistries occur once, and nine repeat across 21 records. Processing differs within eight of those nine groups, so these repeated chemistries sample grain size at fixed chemistry under varying processing conditions rather than serving as replicates; the exception is the CBB04/CBB12 pair, which shares both composition and processing (60% cold work, 875 °C, 1 h) and differs by 32 MPa in YS and 4.7 in HV, the only direct estimate of repeatability the dataset contains. The 93-record YS subset contains 81 unique compositions. One composition appears in both CBB and CBC, so LOBO reduces campaign dependence without removing all exact-composition overlap, and LOO retains an identical chemistry in the training set for 21 of the 94 records. Mean d ranges from 14.66 to 211.75 μm , SD_{grain} from 8.32 to 321.06 μm , YS from 151.5 to 544.5 MPa, and HV from 57.6 to 227.5. Because d and SD_{grain} are strongly correlated ($r = 0.801$), we treat the standard deviation as an exploratory measure of distribution width rather than as an independent mechanism. Measurement scatter is a separate quantity: SD_{YS} is recorded for 80 conditions (0.2–98.7 MPa) and SD_{HV} for all 94 (1.43–12.44 HV). These measurement standard deviations are not model inputs because replicate counts and reporting definitions are not consistent across the dataset. The Bayesian optimization campaigns sample design-relevant regions, not a random composition simplex (Figure 3).

Table 2 defines the cumulative S1-S4 input sets used for matched comparisons in Family 4. These labels refer to feature matrices, not model families. Raw composition is represented by eight atomic fractions when appropriate. Identifiable linear models use seven non-Ni fractions. The six Wen alloy descriptors are valence electron concentration (VEC), mixing enthalpy ΔH_{mix} , mixing entropy ΔS_{mix} , Ω , electronegativity mismatch $\Delta\chi$, and atomic-size mismatch δ [22, 25, 26].

A best-practices framework for physics-informed property prediction

Each tier gains predictive capability and incurs a specific, named risk the framework manages

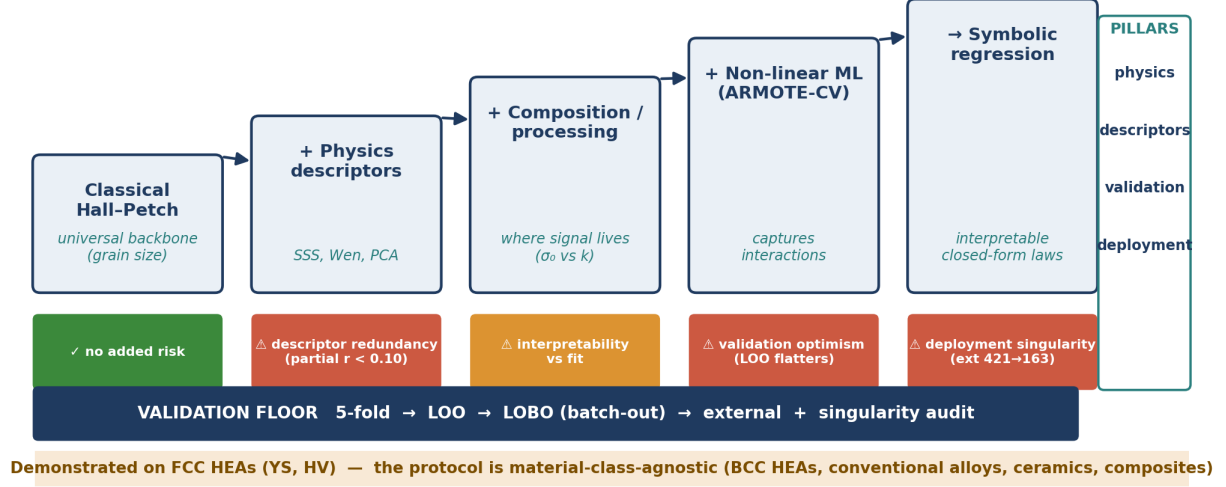


Figure 1: Five family framework used throughout the paper: Classical Hall-Petch, Physics-derived descriptors, Composition/processing, Non-linear estimators, and Symbolic regression. S1–S4 name the matched input sets in Family 4; individual inputs, including SD_{grain} , are also tested in Families 2, 3, and 5. Internal evaluation uses 5-fold CV, LOO, and LOBO; external literature testing and singularity auditing provide additional challenge and equation safety checks.

Table 2: Cumulative input sets used in Family 4.

Set	p	Content
S1	2	$d^{-1/2}$ and SD_{grain}
S2	8	S1 plus six Wen alloy descriptors
S3	11	S2 plus cold work, T_{rx} , and hold time
S4	24	S3 plus composition and SSS-related quantities

3. Methods

3.1. Family 1: Classical Hall-Petch

Equation (1) provides the common grain-size baseline. We also compare nine pre-specified laws: $d^{-1/2}$, d^{-1} , $d^{-1/3}$, $d^{-2/3}$, $\ln(d)/d$, $\ln d$, two composite power-law forms, and a fitted free exponent. AIC, AICc, BIC, and Bayesian PSIS-LOO are used to compare these closely related parameterizations [27, 28, 29, 30]. We apply the same grain-size analysis separately to YS and HV.

3.2. Family 2: Physics-derived descriptors

Family 2 tests whether compact physics-based representations explain YS and HV beyond the grain-size baseline. The Varvenne–Leyson–Curtin (VLC) model uses solute misfit volumes to estimate the zero temperature critical resolved shear stress,

$$\tau_{y0} = 0.051\alpha^{-1/3}\bar{\mu}\left(\frac{1+\bar{\nu}}{1-\bar{\nu}}\right)^{4/3}f_1\left(\frac{\sum_i c_i\Delta V_i^2}{b^6}\right)^{2/3}, \quad (2)$$

followed by a thermal activation correction and a Taylor factor. Here, c_i and ΔV_i are the elemental fractions and misfit volumes, and $\bar{\mu}$, $\bar{\nu}$, and b are composition averaged elastic and

lattice quantities. Weighted Labusch and Toda-Caraballo estimates provide two related SSS representations. The Labusch and Toda-Caraballo constants are calibrated once to the reported CoCrFeMnNi friction stress; VLC is left uncalibrated. The full expressions and parameter values are given in the Supplementary Information.

These implementations use Vegard law volumes and rule-of-mixtures elastic properties [18, 19, 20, 21]. We therefore keep calibrated and uncalibrated results separate. We also distinguish standalone prediction, improvement beyond Hall-Petch, and partial correlation after removing raw composition trends. These tests ask different questions, and partial correlation does not establish any form of independence.

The descriptor branch uses six Wen alloy descriptors together with $d^{-1/2}$, SD_{grain} , and three processing variables. Standardization and six-component PCA are fit again within each training fold before OLS. Full-data loadings are descriptive only. This fold-contained implementation corrects the preprocessing leakage in the earlier analysis.

3.3. Family 3: Composition/processing

Family 3 adds measured composition and processing to the Hall-Petch baseline through a pre-specified OLS hierarchy. Its principal model, M3, is

$$\sigma_y = \beta_0 + \sum_{e \neq \text{Ni}} \beta_e c_e + k_{\text{HP}} d^{-1/2}, \quad (3)$$

where Ni is the composition reference. Other models allow only V in the intercept, composition-dependent slope terms, variation in both intercept and slope, descriptor substitutions, or recorded processing variables. The family name includes processing

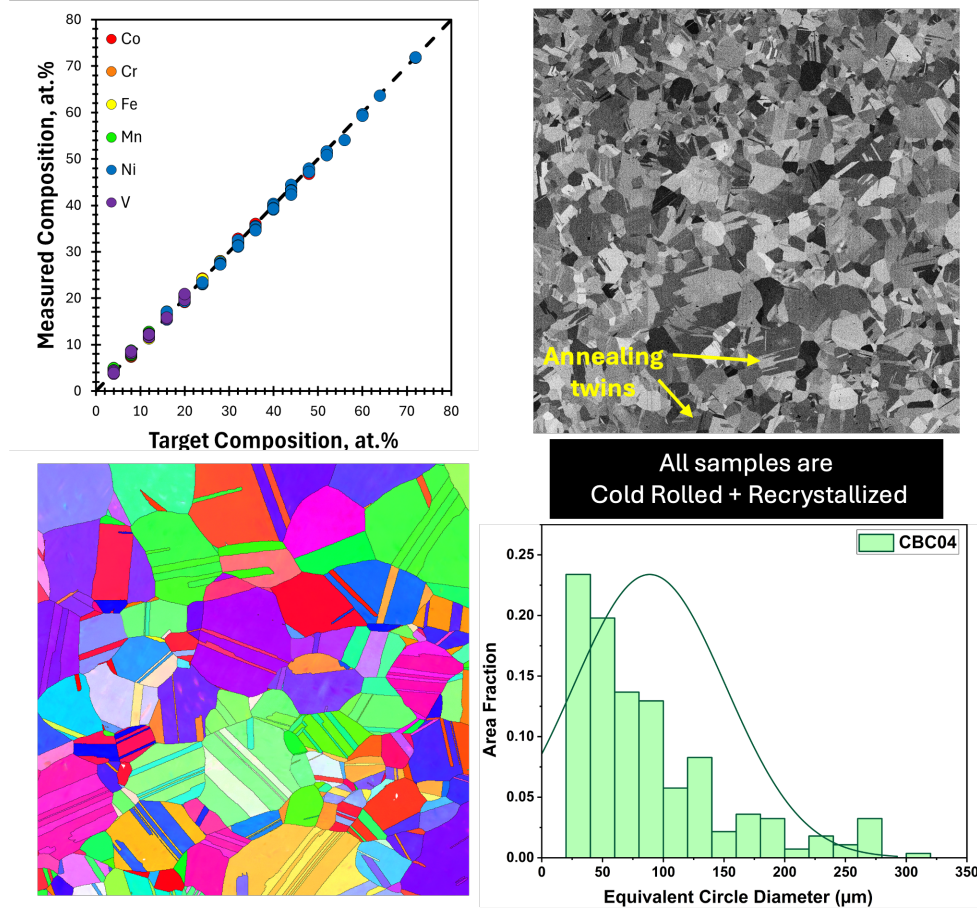


Figure 2: Representative candidate alloy verification: measured and target composition, backscattered-electron imaging, an EBSD inverse-pole-figure map, and the corresponding equivalent-circle grain-size distribution. The right-skewed distribution motivates retaining both mean grain size and its within-map standard deviation.

because the broader hierarchy tests those covariates, although M3 itself uses composition and d without an explicit processing coefficient. Composition, processing, and grain size are interdependent in these campaigns. The fitted coefficients can therefore absorb covariance among them and are not interpreted as elemental constants.

The hierarchy is compared using information criteria and pooled out-of-fold predictions. We test SD_{grain} first as an additive microstructure term and then through M15,

$$\sigma_y = \beta_0 + \sum_{e \neq \text{Ni}} \beta_e c_e + k_{\text{HP}} d^{-1/2} + \gamma SD_{\text{grain}} + \eta SD_{\text{grain}} d^{-1/2}. \quad (4)$$

The interaction allows the grain-distribution width to modify the fitted grain-size contribution. It is pre-specified OLS and can be reproduced from the prepared dataset, but its coefficients remain associational because d , SD_{grain} , composition, and processing are correlated.

3.4. Family 4: Non-linear estimators

Family 4 compares linear controls and non-linear estimators using the same S1–S4 matrices and validation splits. OLS, Ridge, Lasso, and ElasticNet are retained so that the added value of non-linearity can be measured at fixed inputs. The primary audit uses declared, fixed parameter settings with no data-driven

hyperparameter selection. For example, Lasso uses $\alpha = 1.0$, while LightGBM uses 400 estimators, a maximum depth of 3, a learning rate of 0.05, and a feature fraction of 0.8. These are settings chosen for this study, not “library defaults.” An archived ARMOTE-CV table reports nested Optuna tuning for eight regressors. Because the repository does not contain its raw fold objects or generator, we use it only as secondary evidence. We also exclude the legacy full-data-tuned leaderboard from confirmatory rankings.

3.5. Family 5: Symbolic regression

PySR searches three feature pools and three operator sets, while SISSO provides a sparse operator comparison [31, 32]. PySR pool P1 contains grain statistics, P2 adds raw composition and processing, and P3 adds Wen descriptors and processing. The operator sets progress from addition and subtraction to multiplication, division, and protected non-linear operators. SISSO starts from 27 composition, thermodynamic, SSS, and grain-size features and builds two algebraic expansion levels before sparse selection.

In the stored workflow, equations were selected after the complete data search fronts had been viewed. Their constants were then refit within the CV folds. We therefore call these results “post-selection fixed-form CV,” rather than unbiased outer-loop

(a) Composition coverage by element and batch

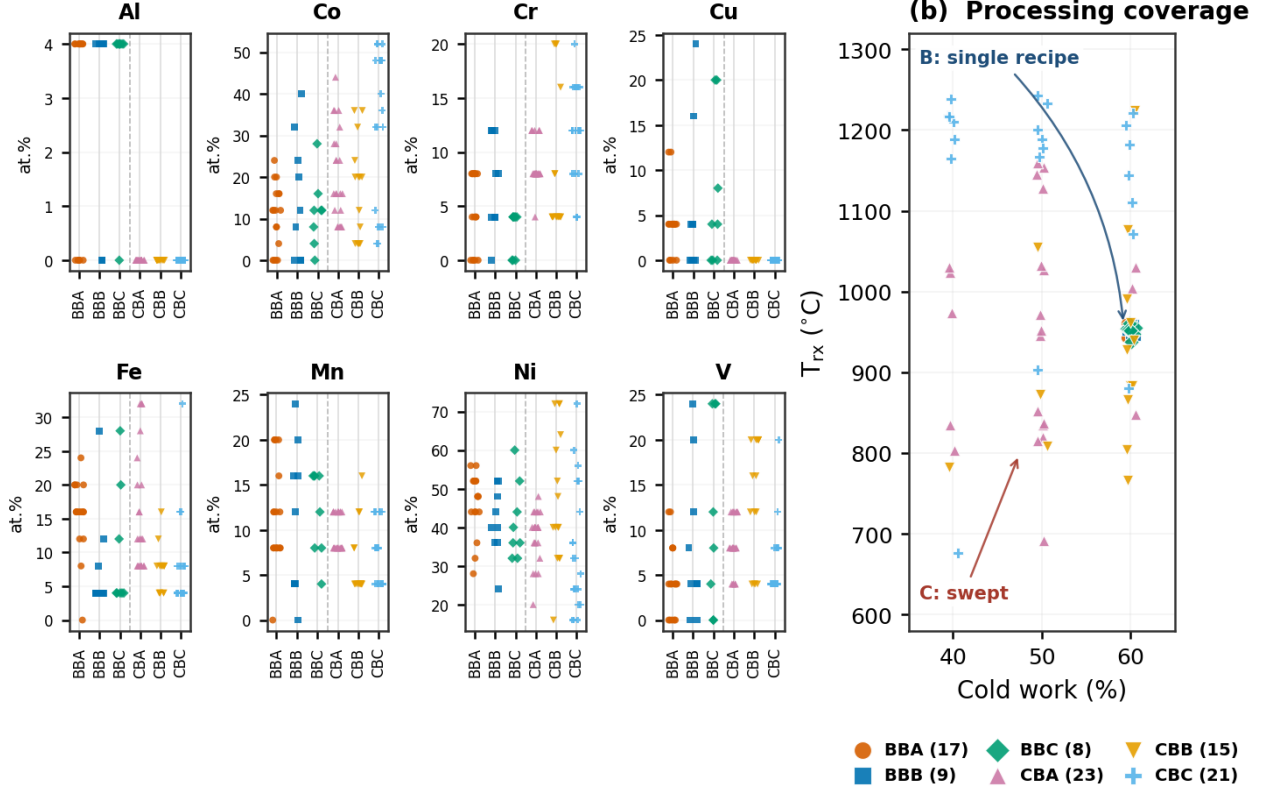


Figure 3: Composition and processing coverage by batch. The campaigns overlap but do not cover identical regions, and processing is partly structured by campaign. This data provenance motivates reporting LOBO along with record level validation instead of relying only on random splits.

estimates of symbolic discovery. Protected square root and logarithm are evaluated as $\sqrt{|x|}$ and $\log(|x| + \epsilon)$. We check each reported closed form for small or transformed denominators, sign changes, non-finite values, and extreme predictions over the observed internal and literature ranges. This audit is equation-specific and does not establish safety outside the stated descriptor envelope. BIC values from symbolic search are not compared directly with the pre-specified linear models because the stored symbolic parameter count omits the cost of searching variables, operators, and structures.

For HV, the selected structure is

$$H_V = c_0 + c_1 \frac{d_0 - d}{SD_{\text{grain}}} + c_2 \frac{\Delta H_{\text{mix}}}{t_{\text{hold}}^2}. \quad (5)$$

The constants are refit in every fold. In the reported equation, d , d_0 , and SD_{grain} are in μm , t_{hold} is in h, and ΔH_{mix} is in kJ mol^{-1} . Thus c_0 and c_1 have units of HV, while c_2 has units of $\text{HV} \cdot \text{h}^2 \cdot \text{mol} \text{kJ}^{-1}$. This is an empirical expression, not a dimensionally universal relation. Its domain excludes $SD_{\text{grain}} \rightarrow 0$ and $t_{\text{hold}} \rightarrow 0$.

3.6. Validation hierarchy and score uncertainty

We use three internal protocols that ask progressively different generalization questions: (i) shuffled 5-fold CV with seed 42; (ii) LOO; and (iii) LOBO across the six BO batches. LOO

tests record-level interpolation, but another record with the same composition can remain in the training set. We therefore do not interpret LOO as strict new composition validation. LOBO tests transfer to a new iteration while retaining data from the broader experimental program and, in one case, the same composition.

For every protocol, predictions from all held-out records are pooled before computing

$$Q^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \quad (6)$$

We report this pooled quantity as cross-validated R^2 . To calculate batch-clustered percentile intervals, we resample the six batches with replacement 1000 times while keeping every batch intact. These intervals show sensitivity to the sampled batches. With only six clusters, they should not be read as precise inferential confidence intervals. They quantify score uncertainty, not per-alloy predictive uncertainty; predictive intervals are outside the scope of this study.

The stored literature compilation has three evidence classes. E1 contains 54 direct strength records: 48 from the Borg/Citrine compilation [33] and six compression records for CrFeNi from Schneider and Laplanche [34]. These records represent only 18 exact compositions and mix sources and test modes. The legacy extraction limited grain size to 3–500 μm , removed YS values above 800 MPa, and omitted one large-grain CrFeNi record

because it departed from the expected Hall-Petch trend. Since two of these rules use the outcome or its expected relationship, we report E1 only as a heterogeneous literature stress test, not as an unbiased external validation set. E2 contains three values reconstructed from published Hall-Petch fits [4]. E3 contains 25 YS proxies converted from reported hardness [35]. We keep E2 and E3 separate from direct measurements.

For the hardness-strength comparison, hardness is converted to stress units and summarized as $C_{\text{eff}} = \text{HV}(\text{MPa})/\sigma_y$. Under the conditional Tabor-Hollomon approximation,

$$n_{\text{app}} = \frac{\ln(C_{\text{eff}}/3)}{\ln(0.08/0.002)} = \frac{\ln(C_{\text{eff}}/3)}{\ln 40}. \quad (7)$$

This is an apparent index, not a measured tensile hardening exponent, because offset YS is not the flow stress at exactly 0.002 true plastic strain and Hollomon behavior is assumed between 0.2% and 8% strain.

4. Results

Table 3 gives the common comparison across model families. Five-fold CV and LOO mainly test interpolation within the pooled dataset, whereas LOBO asks whether a model transfers to an unseen experimental batch. We therefore base the main conclusions on LOBO together with the change from record-level to batch-level validation, rather than on the highest score in any one column. A negative R^2 means that the held-out predictions are worse than using the held-out mean.

The grouped scores should also be read in light of composition coverage. Six principal components represent 96.9% of the composition variance, and the mean pairwise batch-hull overlap in the PC1–PC2 projection is only about 0.21. CBC is relatively isolated in this projection, whereas CBA is well covered by the other batches. LOBO therefore combines easier interpolation-like folds with harder extrapolation folds; the pooled score is not a uniform measure of difficulty.

We do not report a single predictive ceiling. An in-sample composition-mean calculation gives $R^2 \approx 0.86$ if all 72 singleton compositions in the 93-record YS subset are assigned zero residual. In contrast, a calculation based only on repeated compositions gives a formal value near -0.05 because those groups also vary in processing and grain size. A separate estimate near 0.97 treats reported within-alloy standard deviations as irreducible target error, but replicate counts are unavailable. None of these quantities is a defensible out-of-sample ceiling.

4.1. Family 1: Classical Hall-Petch

The classical Hall-Petch law provides the grain-size-only baseline. For YS, it gives training $R^2 = 0.429$, LOO $R^2 = 0.406$ with RMSE = 62.5 MPa, and LOBO $R^2 = 0.373$ (Fig. 4). Its similar LOO and LOBO scores show that mean grain size contains a modest, reproducible YS signal.

The exact grain-size law is less certain. Nine pre-specified forms were compared. Five two-parameter monotonic laws lie within $\Delta\text{BIC} < 2$ and have LOO R^2 from 0.395 to 0.406

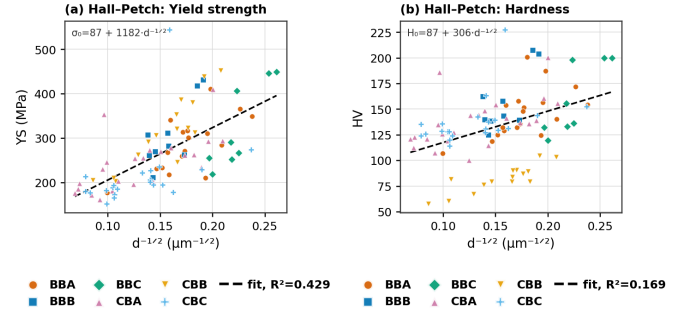


Figure 4: Family 1 Classical Hall-Petch fits for YS and HV. Grain size provides a useful baseline for YS, but mean grain size alone does not transfer for HV.

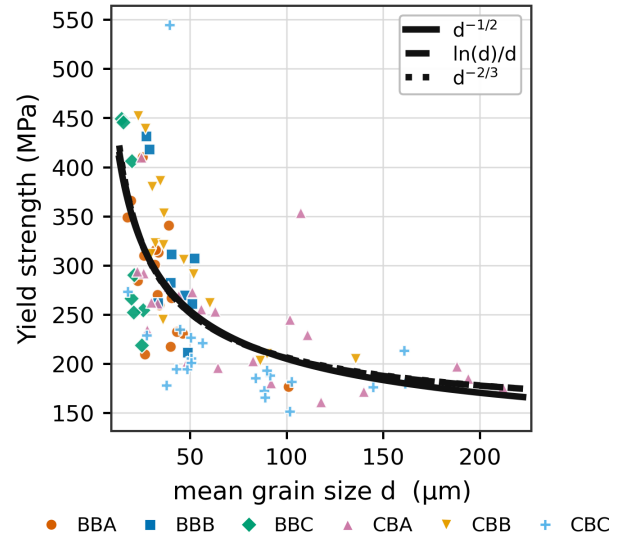


Figure 5: The three best-supported Family 1 grain-size laws fitted to YS over the observed range, with points coloured by batch. The curves overlap closely across the sampled grain sizes, which is why the data do not identify a unique exponent. The corresponding hardness comparison is given in the Supplementary Information.

(Fig. 5). The free-exponent fit gives $n = 0.548$, but its additional parameter raises ΔBIC to 4.5. The Bayesian estimate is $n = 0.53 \pm 0.16$, with a 94% highest-density interval of about $[0.22, 0.82]$. Bayesian stacking places all weight on the classical form, but the candidate laws have nearly equal expected log predictive density, so this boundary weight is not decisive. The data support $d^{-1/2}$ as a simple baseline, not as a uniquely identified exponent or mechanism.

The same baseline does not describe HV well. The classical form has training/LOO $R^2 = 0.169/0.136$ and becomes negative under LOBO (-0.077). A full-data optimized exponent of 1.728 raises LOO R^2 only to 0.166 and is descriptive because the exponent was selected before CV. Mean grain size alone therefore does not transfer as a hardness model.

4.2. Family 2: Physics-derived descriptors

Family 2 tests whether physics-derived descriptors add information beyond grain size and composition. For the

Table 3: Pooled out-of-fold R^2 across the three internal validation protocols. LOBO provides the primary grouped transfer test. Family 4 includes linear controls to isolate the value of non-linearity. The Family 5 HV result is conditional on a structure selected using the full dataset.

Target	Family/model	Evidence status	5-fold	LOO	LOBO
YS	F1 Classical HP	baseline	0.405	0.406	0.373
YS	F2 PCA-OLS	fold-contained	0.462	0.480	0.362
YS	F3 M3	pre-specified linear	0.666	0.652	0.625
YS	F3 M15 SD interaction	pre-specified linear	0.731	0.694	0.694
YS	F4 Lasso S2	fixed settings	0.683	0.674	0.621
YS	F4 LightGBM S2	fixed settings	0.632	0.574	0.615
HV	F1 Classical HP	baseline	0.086	0.136	-0.077
HV	F4 LightGBM S1	fixed settings	0.195	0.323	0.112
HV	F5 fixed form	post-selected	0.725	0.727	0.636

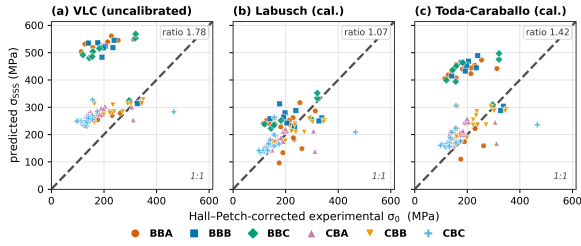


Figure 6: Family 2 SSS predictions compared with Hall–Petch-corrected YS. Calibration can reduce average bias without showing that the descriptor transfers or adds information beyond composition.

SSS estimates, Cantor-anchored Labusch calibration gives a mean prediction-to-observation ratio of 1.07. Calibrated Toda–Caraballo and VLC overpredict by 42% and 78%, and uncalibrated variants overpredict by factors of 2–5. Used alone, VLC, Labusch, and Toda–Caraballo give LOO $R^2 = 0.030, 0.274$, and 0.091 . Adding $d^{-1/2}$ changes these values to $0.396, 0.469$, and 0.399 , compared with 0.406 for Hall–Petch alone (Table 4). Only calibrated Labusch gives a small improvement over that baseline (Fig. 6).

The descriptors do not add information once composition is included. Adding an SSS estimate to a composition model changes LOO R^2 by at most 0.002 , and all linear partial correlations after removing composition are below 0.10 in magnitude. This redundancy applies to the present descriptor inputs; it does not distinguish limitations of SSS theory from uncertainty in atomic volumes, elastic properties, calibration, or unmeasured microstructure.

The broader Wen/PCA representation gives 5-fold/LOO/LOBO $R^2 = 0.462/0.480/0.362$ when PCA is fitted inside each fold. Global preprocessing had produced $0.525/0.516/0.441$, showing that even unsupervised scaling and PCA can inflate held-out scores when fitted before the split. The corrected LOBO result is below the classical Hall–Petch baseline, so the PCA model does not provide a transferable improvement at the batch level. Full-data loadings are therefore used only to describe the descriptor space, not as causal feature importance.

4.3. Family 3: Composition/processing

Family 3 adds composition, processing, and measured microstructure terms to the Hall–Petch baseline. M3 gives

LOO/LOBO $R^2 = 0.652/0.625$ and LOBO RMSE = 49.7 MPa. Its batch-bootstrap interval for LOBO R^2 is $[0.501, 0.670]$, while the six individual batch-held-out scores range from 0.306 to 0.697 . This spread shows that transfer difficulty differs across batches. M3 therefore retains useful YS information beyond mean grain size across the sampled experimental iterations.

The grain-size distribution adds information only in a specific form (Fig. 7). Adding SD_{grain} to M3 as an independent linear term raises LOO R^2 from 0.652 to 0.668 but lowers LOBO from 0.625 to 0.595 . M15, which includes both the additive term and $SD_{\text{grain}}d^{-1/2}$, gives 5-fold/LOO/LOBO $R^2 = 0.731/0.694/0.694$ and LOBO RMSE = 44.9 MPa. Its LOO and LOBO batch-bootstrap intervals are $[0.578, 0.766]$ and $[0.599, 0.733]$. Relative to M3, M15 lowers BIC by 20.1 ; the interaction coefficient has $t = 4.71$ ($p = 1.0 \times 10^{-5}$) in the complete-data OLS fit. M15 is therefore the strongest verified low-complexity YS model when SD_{grain} is measured. The additive control shows that the gain cannot be credited to grain-width measurement alone. Because the term is correlated with mean grain size and processing, it is interpreted as a predictive interaction, not as proof that distribution width changes the Hall–Petch slope.

The individual M3 LOBO scores are 0.697 for BBA, 0.306 for BBB, 0.656 for BBC, 0.423 for CBA, 0.584 for CBB, and 0.409 for CBC; the corresponding RMSE values range from 32.9 to 60.4 MPa. CBC has the lowest projected hull containment (0.476), while CBA has complete containment in the PC1–PC2 projection. Coverage is not the only source of error, but these folds show why the pooled LOBO score should be accompanied by batch-level results.

The complete-data M3 fit gives $k_{\text{HP}} = 766 \text{ MPa}\cdot\mu\text{m}^{1/2}$. Adding the composition-dependent intercept lowers the slope from about $1182 \text{ MPa}\cdot\mu\text{m}^{1/2}$ in the grain-only fit, consistent with composition–grain-size confounding in Family 1. The M3 value is higher than several representative directly reported values, which span 110 – $538 \text{ MPa}\cdot\mu\text{m}^{1/2}$ in the cited pure metals and FCC MPEAs [3, 5, 4, 36]. This comparison is not an exhaustive literature bound, and the fitted M3 value should not be treated as a universal FCC value. Among the composition coefficients (Table 5), V is the largest positive term relative to Ni ($+291$ MPa), but all coefficients are conditional on the sampled compositions and processing routes.

Two sensitivity checks address numerical stability of this fit. Across 1000 grain-size perturbations, the V, Co, Cr, and Fe coefficients change by less than about 15% , whereas k_{HP}

Table 4: Family 2 YS descriptor results. Ratio is mean SSS prediction divided by mean Hall–Petch-corrected YS. Raw R^2 and Spearman ρ compare the complete-data SSS estimates with the corrected YS. Standalone and +HP are LOO results; +HP is Ridge using the SSS estimate and $d^{-1/2}$.

Model	Mean (MPa)	Ratio	Raw R^2	Spearman ρ	Standalone R^2	+HP R^2	+HP RMSE (MPa)
VLC	342	1.78	−6.92	0.53	0.030	0.396	63.1
Labusch, calibrated	206	1.07	0.14	0.55	0.274	0.469	59.1
Toda–Caraballo, calibrated	274	1.42	−3.18	0.49	0.091	0.399	62.9
HP only	—	—	—	—	—	0.406	62.5

Table 5: Family 3 M3 coefficients relative to Ni. Intervals are conditional on the OLS specification and sampled campaigns.

Parameter	Estimate (MPa)	95% low	95% high
Intercept	229.6	152.4	306.7
Al	−201.3	−955.4	552.8
Co	−187.2	−268.1	−106.3
Cr	−307.7	−588.5	−26.9
Cu	−334.0	−651.9	−16.2
Fe	−360.4	−520.1	−200.6
Mn	82.3	−117.6	282.2
V	291.3	41.0	541.5
k_{HP}	765.8	501.5	1030.0

changes by about 39%. Al is not included in the 15% statement because it spans only 0–4 at.% and has a much wider interval. A 10,000-replicate row bootstrap gives standard-error ratios of 0.99–1.54 relative to OLS. These checks support stability within the sampled records, but a row bootstrap does not reproduce campaign-level sampling and cannot establish transfer.

M3 assumes a shared Hall–Petch slope, but the dataset cannot test that assumption decisively. Only nine compositions are repeated, most have two records, and processing changes within some groups. Their descriptive local slopes range from about −8539 to 9282 MPa· $\mu\text{m}^{1/2}$. A fixed-effect summary gives a slope near 2144, Cochran $Q \approx 251$, and $I^2 = 96.8\%$, but this apparent heterogeneity cannot be assigned to composition under the present design. The data therefore do not show that k_{HP} is composition-independent. The supported result is narrower: a composition-dependent intercept with a shared grain-size term predicts YS well across the sampled batches. The physical division among friction stress, processing, and grain-boundary strengthening remains unresolved.

4.4. Family 4: Non-linear estimators

Family 4 tests whether non-linear estimators improve prediction when the input set is held fixed. At S2, Lasso and LightGBM give nearly identical YS LOBO scores of 0.621 and 0.615. Their batch-bootstrap intervals also overlap: [0.516, 0.664] for Lasso and [0.474, 0.664] for LightGBM. This does not prove statistical equivalence, but it gives no evidence that non-linearity improves batch-level transfer at matched inputs. At S1, adding SD_{grain} to the linear YS grain-size baseline changes LOO/LOBO R^2 from 0.406/0.373 to 0.451/0.394, a small gain that is much weaker than the M15 interaction result.

The archived nested ARMOTE-CV summary reports Bayesian ridge at S2 with 5-fold/LOO/LOBO $R^2 = 0.685/0.666/0.613$, which supports the same interpretation. The repository does not contain its generator or raw fold objects, and

the available script tunes once on the full dataset. We therefore retain this result as secondary evidence rather than presenting it as independently reproduced nested CV.

The legacy 17-model panel gives a useful warning about model selection. Its XGBoost model reached LOO/LOBO $R^2 = 0.729/0.574$ after tuning and interaction construction on the complete dataset. Those values are not used in the primary ranking. The archived SHAP analysis explains that fitted model, but correlated descriptors and constructed interactions prevent causal interpretation of its feature rankings.

The same panel reported stacking LOBO $R^2 = 0.707$, but its meta-features were record-level LOO predictions from models that could retain other records from the held-out batch. This leakage prevents a grouped-transfer interpretation. In the fitted XGBoost model, SHAP ranks Ω , $\text{Mn} \times d^{-1}$, $\text{V} \times d^{-1}$, alternative grain-size transforms, and SSS descriptors highly. These attributions describe the selected predictor; they do not isolate mechanisms or prove a composition-dependent Hall–Petch slope.

For HV, the Family 4 S1 matrix contains both $d^{-1/2}$ and SD_{grain} , whereas the Family 1 baseline contains only $d^{-1/2}$. With a linear estimator, this added feature changes LOO/LOBO R^2 from 0.136/−0.077 to 0.232/−0.176: it improves record-level interpolation but worsens batch transfer. LightGBM on S1 gives 0.323/0.112. The comparison therefore changes both the feature set and estimator, and the LOO gain cannot be credited entirely to non-linearity. Descriptor-rich Family 4 input sets are generally negative under LOBO (Fig. 8). The model can interpolate among related records, but it provides little evidence of reliable transfer to an unseen experimental batch.

4.5. Family 5: Symbolic regression

Family 5 searches for compact analytical forms. For YS, the two accuracy-selected PySR equations using composition and processing each reach fixed-form LOO $R^2 = 0.808$ after their constants are refitted. Their archived mean per-batch LOBO scores are 0.696 and 0.689. A descriptor-based accuracy equation gives LOO/stored-LOBO $R^2 = 0.765/0.668$, while the descriptor-based elbow equation gives 0.674/0.555. These archived LOBO values are means of fold-wise R^2 and are not directly comparable with the pooled Q^2 values in Table 3. The descriptor-based accuracy equation contains $SD_{\text{grain}} / \sqrt{|\Delta S_{\text{mix}} - 8.2739|}$. The internal minimum $\Delta S_{\text{mix}} = 8.3637$ is only 0.0898 above its pole. The protected square root keeps the expression real on either side, but does not remove the denominator singularity. This equation is therefore unsafe outside the observed descriptor envelope.

All of these PySR values are post-selection scores: the equation structures were chosen from search fronts built with the full

Composition-dependent Hall-Petch hierarchy for yield strength. Negative ΔBIC favours the model over M3.

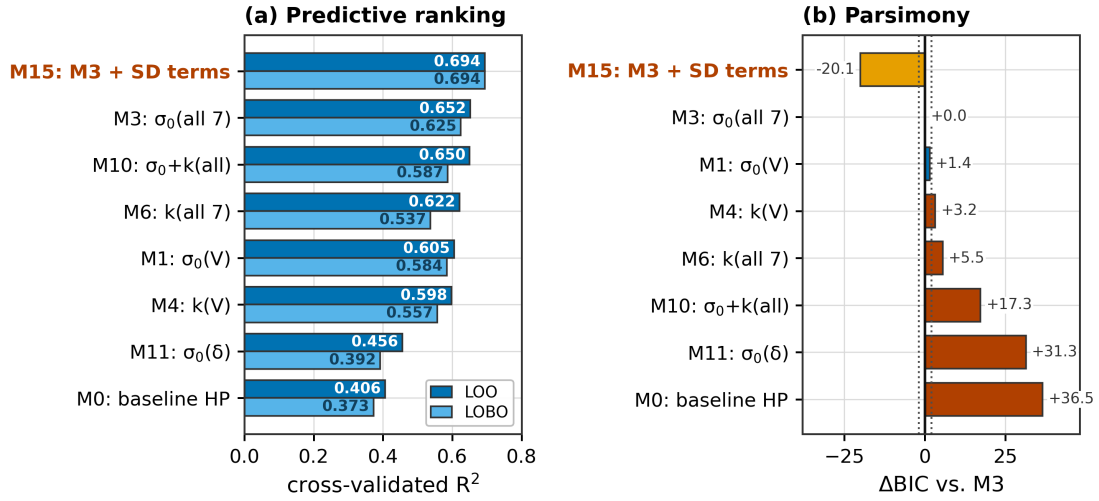


Figure 7: Family 3 Hall-Petch hierarchy for YS. M15 adds SD_{grain} to M3 both directly and through the interaction $\text{SD}_{\text{grain}}d^{-1/2}$. It gives the strongest internal result, but its coefficients remain conditional on the sampled campaigns and correlated microstructure.

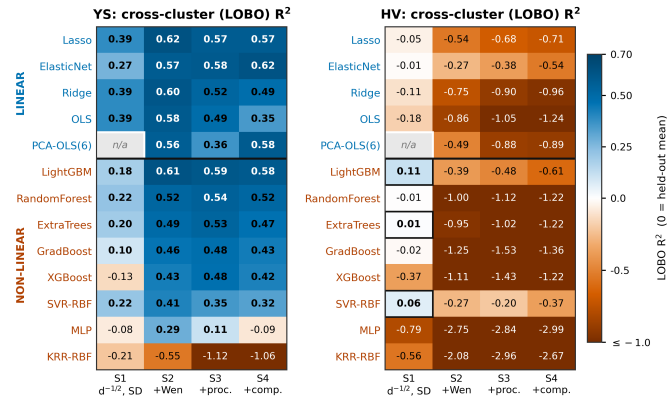


Figure 8: Family 4 matched-input LOBO comparison. For YS, the linear control is competitive with the non-linear estimators. For HV, many descriptor-rich combinations are negative under batch-level validation.

dataset. They are not nested estimates of symbolic-regression performance and cannot be ranked directly against the pre-specified models. The complexity-loss front from which they were taken is shown in Fig. 9. PySR BIC also counts fitted constants but not the cost of searching variables, operators, and structures, so it is not compared with the BIC of the pre-specified linear hierarchy. The SISSO equations have the same selection limitation. Grain-width terms recur across searches and are useful hypotheses, but not independent evidence of a mechanism: all searches use the same 94 records, and d is correlated with SD_{grain} .

Refitting the selected HV structure gives

$$H_V = 222.3 + 84.5 \frac{6.93 \mu\text{m} - d}{\text{SD}_{\text{grain}}} - 1.005 \frac{\Delta H_{\text{mix}}}{t_{\text{hold}}^2}. \quad (8)$$

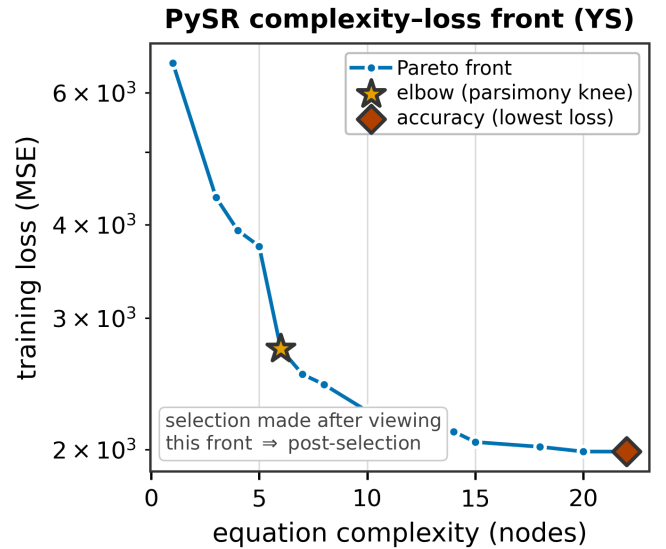


Figure 9: Family 5 PySR complexity-loss front. Choosing an elbow or accuracy equation after viewing this front places structure selection outside the later constant-refit CV, so the result must be labeled post-selection.

The three coefficient bootstrap intervals are $[202.9, 240.1]$, $[69.4, 98.3]$, and $[-1.218, -0.803]$. The corresponding bootstrap means and standard deviations are 221.82 ± 9.64 , 84.12 ± 7.34 , and -1.006 ± 0.106 ; each interval excludes zero, conditional on the selected structure. The full-data fit gives $R^2 = 0.744$ and $\text{RMSE} = 16.90$ HV. With the structure fixed and only its constants refitted, 5-fold/LOO/LOBO $R^2 = 0.725/0.727/0.636$, with $\text{RMSE} = 17.50/17.46/20.16$ HV. The batch-bootstrap interval for LOBO R^2 is $[0.329, 0.765]$.

The equation retains useful batch-held-out performance, but its structure was selected after viewing the full-data search fronts.

The zero at $d = 6.93 \mu\text{m}$ is not a singularity; the unsafe limits are $\text{SD}_{\text{grain}} = 0$ and $t_{\text{hold}} = 0$. The observed data remain away from both limits. The two non-constant terms have different provenance risks. Hold time is fixed at 0.5 h for all 35 B-campaign records and takes values of 1, 2, 4, or 8 h in Campaign C, so hold time separates campaign with $\text{AUC} = 1.000$ and cannot be identified within Campaign B. The combined $\Delta H_{\text{mix}}/t_{\text{hold}}^2$ term also separates campaign strongly ($\text{AUC} = 0.970$). In contrast, $(6.93 - d)/\text{SD}_{\text{grain}}$ is nearly campaign-neutral ($\text{AUC} = 0.533$). The hold-time term must therefore be treated as a campaign proxy in these data; the grain-width term remains a post-selected, correlated microstructure hypothesis. Testing either term as a mechanism requires a prospectively frozen equation and direct distribution descriptors such as SD/d , lognormal width, or $E[d^{-1/2}]$.

The SISSO comparison also separates numerical safety from predictive accuracy. SISSO Full places δ_μ in a denominator and becomes unstable when constituent shear moduli are similar. Removing that term lowers the aggregate literature RMSE from 420.6 to 162.9 MPa. The bounded form is numerically safer, but its aggregate $R^2 = -1.37$ is still unsuitable for prediction.

Additional symbolic controls give lower or unstable results. An expanded SISSO search reports LOO/LOBO $R^2 = 0.604/0.350$ when the selected transforms change across folds. A universal-operator EML control gives LOO $R^2 = 0.321$ at depth one and becomes unstable at greater depth. A recalibrated pure-metal descriptor relation gives LOO $R^2 = 0.346$. These controls suggest that restricted operators can provide useful inductive bias, but the archived files do not establish that every structure-selection step was nested.

4.6. Hardness–yield correspondence

The different YS and HV results motivate a direct comparison of the two properties. Across the 93 paired records, $C_{\text{eff}} = 5.13 \pm 1.36$, corresponding through Eq. (7) to $n_{\text{app}} \approx 0.15 \pm 0.07$ after propagating the reported C_{eff} summary. The global Spearman correlation is 0.46, although the within-batch correlations range from 0.70 to 0.95. Pooling weakens, but does not reverse, the association; this is aggregation-induced attenuation rather than a strict Simpson paradox. The partial rank correlation after controlling for d is only 0.24.

The archived descriptive relation is $\log C_{\text{eff}} = 1.36 + 0.10 \log d - 2.06 x_V$, with d entered in the fitted numerical units. It is not dimensionally invariant and has not been prospectively validated. Ranking agreement also depends on the cutoff (Fig. 10): 2 of the 5 hardest alloys are among the 5 strongest, increasing to 8/10 and 11/20 for broader sets. HV may help rank alloys within a narrow composition and processing window, but the results do not support a single transferable HV-to-YS conversion or identify V as a causal source of the pooled attenuation.

4.7. Literature stress test

The E1 literature set provides a final stress test outside the experimental program. On its 54 direct-strength records, M3 has the lowest RMSE, 149.5 MPa, but $R^2 = -0.508$. SISSO Robust gives RMSE = 160.0 MPa and $R^2 = -0.726$, while

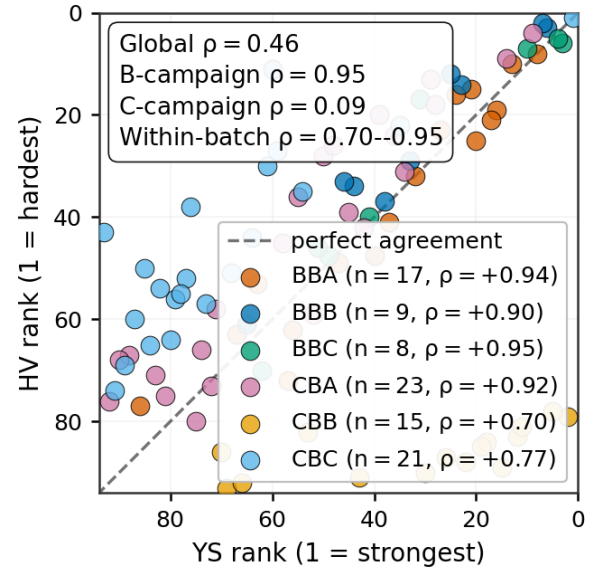


Figure 10: HV rank versus YS rank by batch. Strong within-batch ordering becomes weaker after batches are pooled because their composition and processing coverage differs.

Table 6: Legacy E1 direct-strength literature stress test. Negative R^2 indicates performance worse than predicting the E1 mean.

Model	R^2	RMSE (MPa)	Bias (MPa)
F3 M3	-0.508	149.5	-76.6
F5 SISSO Robust	-0.726	160.0	-61.3
F5 SISSO Full	-5.084	300.3	+50.0

the small-denominator term in SISSO Full contributes to $R^2 = -5.084$. All three models are worse than predicting the E1 mean (Table 6). The test is useful for exposing domain sensitivity and unsafe equations, but not for estimating unbiased external performance: E1 mixes sources and test modes and was filtered using outcomes.

The other literature classes give a different appearance because they are not independent direct measurements. On the three E2 values reconstructed from published Hall–Petch fits, M3 and SISSO Robust give descriptive $R^2 = 0.677$ and 0.849 , with RMSE = 49.0 and 33.5 MPa. With $n = 3$, these R^2 values are not interpretable as generalization estimates. On the 25 E3 hardness-converted proxies, the same models give $R^2 = -2.157$ and -9.642 . Pooling E1–E3 gives M3, SISSO Robust, and SISSO Full $R^2 = -0.580$, -1.373 , and -14.813 , respectively. Across all 18 stored PySR YS variants, the pooled E1–E3 R^2 is also negative; the best is -0.031 (RMSE = 107.4 MPa), and the highlighted descriptor accuracy equation gives -0.295 (RMSE = 120.4 MPa). These pooled values are reported only as numerical stress tests; they do not increase the amount of direct external evidence.

5. Discussion

5.1. Yield strength across the five families

The YS results form a clear progression. Family 1 establishes a useful grain-size baseline, but several nearby exponents fit almost equally well. Family 2 shows that the tested SSS and Wen/PCA representations do not provide a transferable improvement with the available approximations. Family 3 supplies the largest verified gain: M3 combines a composition-dependent intercept with a shared Hall–Petch slope, while M15 adds a grain-width interaction and improves LOBO from 0.625 to 0.694. The additive SD_{grain} control does not transfer as well, so the result is specific to the interaction. Family 4 reaches similar batch-held-out accuracy with a linear control and a non-linear model, so estimator flexibility is not the source of the gain. Family 5 produces compact equations and further grain-width hypotheses, but its equation search was not nested and its literature performance is negative. M15 is the strongest verified low-complexity YS model when grain-distribution width is measured; M3 remains the simpler choice when only mean grain size is available. Neither is a universal Hall–Petch law.

5.2. Hardness requires a different model

The same sequence gives a different result for HV. Mean grain size, physics descriptors, composition, and the tested non-linear models all transfer poorly across batches. Adding SD_{grain} linearly improves LOO but worsens LOBO. The post-selected Family 5 fixed form performs well under LOBO after coefficient refitting, but its structure selection was not nested. Its grain-width term is nearly campaign-neutral, whereas its hold-time term closely tracks campaign identity. These terms are worth testing separately, but they do not yet define a hardness law. The variable C_{eff} and the rank analysis also show why HV should not be treated as a direct substitute for YS across different composition and processing domains.

5.3. The scientific conclusion depends on validation scale

The same fitted model can support different conclusions at different validation scales. The clearest examples are for HV: classical Hall–Petch changes from LOO $R^2 = 0.136$ to LOBO $R^2 = -0.077$, LightGBM S1 falls from 0.323 to 0.112, and the legacy tuned XGBoost model falls from 0.729 to 0.574. In Family 2, global PCA gives LOO/LOBO $R^2 = 0.516/0.441$, whereas fold-contained PCA gives 0.480/0.362, showing how preprocessing and grouping compound. In contrast, M3 changes only modestly from 0.652 to 0.625, and M15 gives 0.694 under both protocols. The Family 5 HV fixed form retains $R^2 = 0.636$ across batches, but this estimate remains conditional on a structure selected from the full dataset and includes a hold-time term confounded with campaign. LOBO is a useful internal transfer test, but it is not equivalent to deployment because the batches share program-level synthesis, characterization, and design practices.

M15 is therefore the best verified YS model for record-level interpolation and new-batch prediction within the combined experimental program when SD_{grain} is available. M3 is a simpler

alternative that does not require grain-distribution width. Neither model is a universal Hall–Petch law or a strict test on unseen compositions. Lasso is also useful as a matched predictive control, although it has less direct physical structure. The Family 5 equations remain discovery hypotheses because their structure selection was not nested and their literature evidence is unfavorable. For HV, Eq. (8) is the strongest candidate within the sampled batches, but it should be frozen and tested prospectively before it is treated as a transferable hardness law.

5.4. Physical interpretation without overreach

The data support a grain-size baseline for YS together with additional composition- and grain-distribution-associated signals. They do not uniquely attribute those signals to friction stress, solute misfit, boundary chemistry, processing, or hetero-deformation-induced strengthening. The apparent preference for a constant Hall–Petch slope may result from limited identifiability rather than true invariance. SD_{grain} appears in M15, the Family 4 feature sets, and several symbolic expressions, but this recurrence is not independent confirmation because the models use the same records and correlated predictors. Raw standard deviation scales with the mean and can also depend on mapped area, grain count, weighting, and segmentation.

YS and HV are not interchangeable labels. Across the 93 paired records, the effective ratio $HV(\text{MPa})/\sigma_y$ is 5.13 ± 1.36 . Strong within-batch rank correlations become weaker when the campaigns are pooled. A single Tabor conversion cannot remove the effects of work hardening, constraint, processing, and campaign structure. Separate models and validation are needed for each property, even though both measurements arise from related plastic-deformation processes.

5.5. Best practices for physics-informed property ML

The five families are most useful as a sequence of scientific questions, not as a leaderboard. Family 1 defines a physically interpretable baseline. Family 2 tests whether physics-based descriptors explain the remaining error. Family 3 asks what measured composition and processing add. Family 4 tests estimator flexibility using matched inputs and budgets. Family 5 generates closed-form hypotheses and therefore requires the strictest audit of selection and domain.

For all five families, the validation split must match the proposed use. LOO measures record-level interpolation and must be interpreted alongside the repeated-composition audit. LOBO is needed for claims about a new experimental iteration. Preprocessing, feature selection, hyperparameter tuning, and equation discovery must all occur within that validation boundary. Uncertainty estimates should preserve experimental clusters. Direct measurements should remain separate from values reconstructed from equations or converted from other properties. External sets should be defined before outcomes are inspected and should document composition, processing, grain statistics, phase state, temperature, and test mode. Closed forms also require a singularity audit over both observed and intended deployment domains.

Table 7 maps each claim made in this work onto the evidence that supports it and the interpretation that evidence allows. Mechanistic claims require more than agreement among predictive

models. A proposed term should survive alternative parameterizations, dimensional analysis, propagation of measurement uncertainty, and a prospective experiment that varies the proposed cause independently of confounders. For the present grain-width hypothesis, such an experiment should hold mean grain size constant while changing the distribution shape. It should also record complete EBSD distributions and sampling metadata, with the equation frozen before testing.

5.6. Limitations and reproducibility

The analysis includes six batches from two sequential campaigns within one research program. Bayesian-optimization sampling narrows the observed domain, and the composition and processing coverage differs among batches. Exact replicates are limited, processing and composition are not orthogonal, and complete grain-size distributions are reduced to a mean and standard deviation. The literature stress set is not suitable for model selection or an external accuracy claim. A future benchmark should be registered before prediction and should use consistent tensile definitions and complete processing metadata.

The revised repository separates reproducible results from archival, post-selected, and proxy-based results. The scripts `grouped_validation.py` and `sdgrain_models.py` generate the primary grouped-validation table and the M3/M15 grain-width audit from the prepared descriptor dataset. The standard OLS, information-criterion, and grouped-CV results for M15 are independently reproduced here. Its earlier Bayesian stacking weight and PSIS-LOO comparison remain archival because that probabilistic analysis was not regenerated. In addition, the archived ARMOTE-CV summary reports nested tuning, but the available script performs hyperparameter tuning on the complete dataset, so the summary cannot be distinguished from a leaky implementation without its per-fold records. PySR equations were similarly selected on complete-data Pareto fronts; we refit their constants per fold but did not rediscover their structure inside a training fold. Both would require fold-level generators and stored out-of-fold predictions to verify, and we therefore treat both as archival.

6. Conclusions

Physics-informed ML improves materials-property modeling only when physics constrains both the representation and the claim. In the fixed hierarchy of Fig. 1, Family 1 provides the grain-size baseline. Family 2 shows that the tested SSS and Wen/PCA descriptors are redundant or conditional on the campaign. Family 3 gives the clearest verified improvement for YS through a composition-dependent Hall–Petch intercept and a grain-width interaction. Family 4 shows no demonstrated advantage for non-linearity when the inputs are matched. Family 5 produces compact hypotheses, but their selection and transfer remain unresolved.

The key result is the gap between record-level and batch-level validation. M15 gives the strongest verified low-complexity YS performance when SD_{grain} is measured, while M3 retains moderate performance without that feature. The tested non-linear

estimator does not improve on its linear control at matched inputs. The symbolic HV form performs best across the sampled batches but remains conditional on post-selected structure and a hold-time term confounded with campaign. The practical implication is straightforward. Start with the known physics, test compact descriptors, then add measured composition, processing, and microstructure before increasing estimator complexity. Choose validation groups that match the intended decision. A model should not progress from interpolation to physical interpretation or alloy design without fold-contained selection, cluster-preserving score intervals, outcome-independent external data, and prospective mechanistic tests.

CRedit authorship contribution statement

Mrinalini Mulukutla: Methodology, Software, Investigation, Formal analysis, Data curation, Visualization, Project administration, Writing – original draft. **Shakti Prasad Padhy:** Methodology, Software, Formal analysis, Project administration, Visualization, Writing – original draft. **Wenle Xu:** Data curation, Investigation, Methodology, Writing – original draft. **Bibhu Prasad Sahu:** Data curation, Investigation, Methodology, Writing – original draft. **Joydeep Kundu:** Data curation, Investigation, Methodology, Writing – original draft. **Vasanth C. Shunmugasamy:** Data curation, Investigation, Methodology, Writing – original draft. **Douglas Allaire:** Conceptualization, Formal analysis, Funding acquisition, Software, Supervision, Writing – review & editing. **Ibrahim Karaman:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing. **Raymundo Arróyave:** Conceptualization, Formal analysis, Funding acquisition, Software, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 7: Claim–evidence map after grouped-validation and provenance audits.

Claim	Evidence	Supported interpretation
Composition improves YS	M3 LOO/LOBO $R^2 = 0.652/0.625$	Supported for record- and batch-level prediction within the present program
Grain-width interaction improves YS	Additive SD gives 0.668/0.595; M15 gives 0.694/0.694	Supported as a predictive interaction when SD is measured; mechanism not identified
Non-linearity improves YS	Lasso/LightGBM LOBO 0.621/0.615	No demonstrated advantage for the tested non-linear model at matched S2 inputs
The HV closed form generalizes	Fixed-form LOO/LOBO 0.727/0.636; post-selected structure	Strongest internal HV candidate; prospective validation is still required
Grain-width terms identify a mechanism	Shared data, correlated d and SD_{grain} , post-selected forms	Testable hypothesis requiring controlled microstructure and prospective validation
Models deploy to literature alloys	All evaluated E1 stress-test $R^2 < 0$; set has mixed modes and outcome-informed curation	No deployment claim and no unbiased external estimate

Data and code availability

The data generated and analyzed during this study are included in the Supplementary Information and are also available as an Excel database in the GitHub repository below.

All code used in this project, including the ARMOTE-CV non-linear ML modeling, is available at <https://github.com/mrinalinimulukutla/Hall-Petch-Modeling>.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Claude Opus 5 and ChatGPT 5.6 Sol for data analysis assistance and prose editing. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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